American Computer Science League

2023-2024 ● Contest 2: Short Problems ● Intermediate Division

1. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

5 8 − 6 9 6 − * 3 / +

A. 6
B. 11
C. 3
D. -2
E. 18

2. Prefix/Infix/Postfix Notation

Translate the following infix expression into prefix:

$$\frac{(a+b)^2-c}{d+e(b^2-a)}$$

A. /−^++ab2cd*e−^b2a
B. /−^ab2c++d*e−^b2a
C. /^−+ab2c+d*e−^b2a
D. /−^+ab2c−d*e+^b2a
E. /−^+ab2c+d*e−^b2a

3. Bit-String Flicking

Evaluate:

(LCIRC-2 (RSHIFT-1 (NOT 10100))) OR (RCIRC-1 (LSHIFT-2 10111))

A. 01011
B. 00101
C. 01111
D. 11110
E. 11111

4. Bit-String Flicking

Solve the following expression for X if X is a 5-bit string:

(LCIRC-2 01101) AND X OR (RSHIFT-1 01111) = 10111

A. 0*1*0
B. 1****
C. 0***1
D. 01*01
E. 00101

5. LISP

Evaluate the following LISP expression:

(CAR (CDR (CAR (CDR '(5 (2 (3 4) 1) (6 7 8) 9)))))

A. (7 8)
B. (6 7 8)
C. (3 4)
D. ((3 4) 1)
E. (2 (3 4))

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1. 前缀/中缀/后缀

如果下方所有数字都是一位数，请计算下方后缀表达式:

5 8 − 6 9 6 − * 3 / +

A. 6
B. 11
C. 3
D. -2
E. 18

2. 前缀/中缀/后缀

请将下方中缀表达式转换为前缀表达式：

$$\frac{(a+b)^2-c}{d+e(b^2-a)}$$

A. /−^++ab2cd*e−^b2a
B. /−^ab2c++d*e−^b2a
C. /^−+ab2c+d*e−^b2a
D. /−^+ab2c−d*e+^b2a
E. /−^+ab2c+d*e−^b2a

3. 比特位操作

请计算下列表达式：

(LCIRC-2 (RSHIFT-1 (NOT 10100))) OR (RCIRC-1 (LSHIFT-2 10111))

A. 01011
B. 00101
C. 01111
D. 11110
E. 11111

4. 比特位操作

X 是一个 5 位的位串，请根据下方表达式计算 X 的值：

(LCIRC-2 01101) AND X OR (RSHIFT-1 01111) = 10111

A. 0*1*0
B. 1****
C. 0***1
D. 01*01
E. 00101

5. LISP 语言

请计算下方 LISP 表达式:

(CAR (CDR (CAR (CDR '(5 (2 (3 4) 1) (6 7 8) 9)))))

A. (7 8)
B. (6 7 8)
C. (3 4)
D. ((3 4) 1)
E. (2 (3 4))